AMRITA VISHWA VIDYAPEETHAM

23CSE452 – BUSINESS ANALYTICS

Capstone Final Project Report

EV Charging Demand Prediction and Optimization

An integrated analytics approach for workplace EV charging with ACN-Data

Team Number: 8

Team member	Register number
Adithyaraman R	CB.SC.U4CSE23601
Muppidi Chandana	CB.SC.U4CSE23228
D Bhuvaneshwari	CB.SC.U4CSE23612
Khandavilli V V S D Gnanesh	CB.SC.U4CSE23662

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Course Faculty / Guide: Rohini S

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Declaration

Team 8 below hereby declare that this report is our individual work submitted for the course 23CSE452- Business Analytics. We collected the data ourselves from public ACN-Data REST API. We have not used any pre-built dataset available on Kaggle, UCI or any other gitHub repository. We have not used any synthetically generated data. We have appropriately cited and listed references for the ideas, figures, algorithms or any software used from other sources. We have specified our use of any generative-AI assistance (if any) in section 8.3.

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Signatures:

Team member	Signature
Adithyaraman R	
Muppidi Chandana	
D Bhuvaneshwari	
Khandavilli V V S D Gnanesh	

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Abstract

The demand for workplace EV charging is not constant throughout the day, week and across different chargers and policies making capacity planning and peak control challenging for operators of charging networks. This project tackles this challenge with an end-to-end analytics pipeline based on 31,424 charging sessions acquired from ACN-Data Caltech REST API (25 April 2018 -13 September 2021) with 31,387 valid sessions after validation. In the exploratory analysis for Unit 1, 4 operational regimes were observed, 58% demand reduction in the weekdays during paid charging and almost half of the plugged-in time spent idle. In Unit 2, chronological validation for next day session counts, comparing baseline vs machine learning models: 21.2% WAPE (5.4 session MAE) in the paid charging period is the best achievable by Histogram Gradient Boosting, compared to 31.1% by the best baseline model. In Unit 3, two approaches are applied, first SARIMAX forecasting to reduce 28-day-ahead MAE from 3.86 sessions/day to 3.18 sessions/day compared to weekday average mean baseline and second Apriori association rule mining to reduce 35,149 rule candidates to 295 interpretable rules (165 rules for driver behaviour). The results are communicated with a live interactive dashboard. Evidence shows there is no need to add more chargers. Instead, manage peak from 10:30-15:45 and idle parking with idle time policies for the operator.

Executive Summary

Problem. It is challenging to determine the appropriate amount of chargers, when to service them and price/access limitations for the workplace charging site because of the drastic demand shift in free/capped/paid and COVID impact periods.

Main Evidence: From 1 Nov 2018, charging was made to be paid. Weekday session drops by 58%. The demand shows steep morning ramp up. However, charger occupancies have their high point in 13:00-14:00, which is around 5 hours after peak arrival. At most, 51 out of 54 chargers are occupied. Many chargers have large portion (more than half) of the plugged-in time as idle as the vehicles stay much longer than needed for charging.

KPI outcomes: With the chosen model for next day prediction, it outperforms the simple average of history (baseline) by 10 percentage point of errors in paid period. With 8 week SARIMAX forecast, we expect about 24.7 session per day. At most, we expect around 21 vehicles to be at one time with 33 chargers used and 55 installed. For the busiest weekday, we need about 26 chargers for 90% buffer. And if we have pre-COVID demand level, we need about 37 chargers.

Business value: No new hardware needed. Save capital expenditure on chargers. Tackle the real bottleneck which is idle parked vehicles with policy.

Recommend actions: (1) Apply weekly refitted SARIMAX forecast for 1-4 weeks for capacity plan. (2) Have idle fee policy after grace period or move your car notifications in 10:00-16:00. (3) Re-activate idle chargers only when demand increases. Plan maintenance on weekends or closure days and outside 08:00-21:00.

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Abbreviations

Abbreviation	Meaning
ACN	Adaptive Charging Network
EDA	Exploratory data analysis
EV	Electric vehicle
HistGB	Histogram Gradient Boosting
MAE / RMSE	Mean absolute error / root mean squared error
SARIMAX	Seasonal ARIMA with exogenous regressors
WAPE	Weighted absolute percentage error
KPI	Key performance indicator

1. Introduction, Business Objectives, Literature Review

1.1 Scope and context

This study focuses on charging of EVs (electric vehicles) at workplace / campus. The key players are: (1) charging network operator / facility managers who design the charger capacity, maintenance, people and pricing; (2) drivers who use the chargers. The time of day and weekday and station and period policy profiles for charging are skewed. There is a high demand for long-stay vehicles (hours) on chargers. Energy allowances and charged charging. COVID remote operations have caused a shift in demand. The demand shifts complicate the capacity design and peak-period demand control without over-building [3].

Scope. The study looks at a single site (Caltech ACN workplace network). It is session level and daily level. The measures of demand are number of charging sessions/day. Energy delivery, charging time and occupancy are secondary outcomes. It does not include per-hour, per-station forecasting. It does not use weather input nor live data refresh (yet, see Section 9, completed scope).

General objective. Forecasting of EV workplace charging demand in the future and finding the regular charging patterns. For supporting charger capacity design. Peak-period management.

Analytical questions. (i) What are the hour, weekday, station, duration, energy, regime demand profiles? (ii) Number of sessions expected in the next day and next 8 weeks? Number of chargers to cover most days? (iii) What is the recurrent pattern of charging behaviours (arrival, stay, energy, idle time)? Implications on charger allocation and idle time policy?

The measurable KPIs are listed in Table 1. The baseline and target are from results (presented later in this document).

Table 1. Business objectives and KPIs and success criteria.

Business objective	KPI / unit	Baseline → result / target	Unit / method
O1. Understanding demand pattern	Sessions/day by regime. Share plugged in time idle (%)	R1 74.0, R2 88.5, R3 36.8, R4 8.7 weekday sessions/day. Idle share 48% (validated data)	Unit 1 (EDA)
O2a. Next day forecast demand	WAPE(%) MAE (sessions)	baseline 31.1% WAPE / 7.9 session -> histGB 21.2% / 5.4 (paid period)	unit 2
O2b 1-8 week ahead forecast size capacity	MAE (session/day) number of chargers 90% buffer	baseline MAE 3.86 -> SARIMAX 3.18 busiest day of week ~ 26 chargers needed vs 55 installed	unit3 SARIMAX

Business objective	KPI / unit	Baseline → result / target	Unit / method
O3 detect recurrent pattern of behaviour	number of stable non-trivial rules support/confidence/lift	candidate rules 35149 -> rules 295 –threshold s>=2%, c>=60%, lift>=1.2	unit3 apriori

End to end workflow. We're sourcing data from ACN-Data api. Data is validated and prepared. After we're exploring data (unit 1) then using it for predictive modelling of next day demand (unit 2). Finally we send data to two methods in unit 3: SARIMAX for time-series forecasting and Apriori association rules mining. All results are feeding interactive dashboard and also business recommendations. Overview of the flow is in Fig. 1

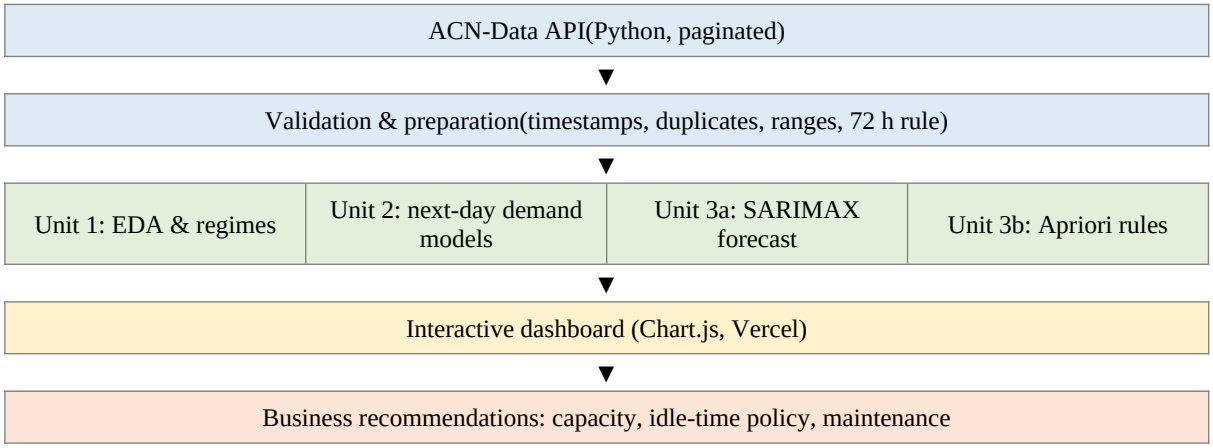


Figure 1. End to end project workflow. Units 1-3 connected with interactive application.

1.2 Related analytics and research gap

There are 3 groups of related works on workplace charging. First is Lee et al. [1] where they released ACN-Data - open data with real charging sessions (Caltech campus & JPL), analysing charging demand, user behaviour (like unreliable users departure time declaration). Second is forecasting studies: to predict charging demand using history of sessions. Third one is studies on scheduling and planning e.g. Lee et al. [2] with Adaptive Charging Network framework which is using same data for workplace charging stations planning, capacity optimization & scheduling.

Limitations and gap. Usually, demand prediction and infrastructure planning are regarded as distinct analytical tasks. Moreover, in many studies, only a single fixed operation period is considered, which means that pricing and access policy shifts are not modeled as structural changes. However, there is an opportunity for a combined (integrated) approach that includes both future demand prediction and regular analysis of the charging behavior. In our project, we apply machine learning and time-series methods for demand prediction, use Apriori [4] to mine

behavior rules, and account for policy regimes explicitly. We will compare our results to the mentioned studies numerically in Section 6.2.

2. Dataset Collection, Validation and Preparation

2.1 Provenance, coverage and challenges

Source and collection. We are gathering the charging sessions from the Caltech workplace network. The charging sessions are collected by the team via the ACN-Data REST API (endpoint /api/v1/sessions/caltech) with Python and API token. Due to the limitation that the API only provides a maximum of 100 records in one request, the data collection is paginated. The dataset is released for research purposes on the ACN-Data platform [1, 7]. We did not leverage any pre-made datasets on Kaggle/UCI/GitHub. Nor did we create any synthetic datasets.

Size and coverage. In the raw data, we have 31,424 sessions and 13 source fields. Upon validation we have 31,387 sessions left. Feature engineering gave us 59 analytical variables. Date range 25 April 2018 -13 September 2021. In terms of days in the calendar -we have 1,238 days (155 weeks and 1,132 days with data). We have a level of observation: charging session level. We want to predict/modelling per day -number of sessions. Coverage of approx. 54-55 charging spaces.

Data collection issues. Only 14,400 out of the 31,424 available records were collected at Review 1 (due to intermittent 502 and connection issues). Collection process was extended and completed. We use the full set of records (2018-2021) in this report. Identified there is a record break where no sessions were recorded between 4 August 2020 and 17 November 2020 (106 days empty). We treated it as missing data -not zero. Note, for only some of the sessions we have user_id and user_input (1,730 at Review 1). Note, doneChargingTime is not provided for 2,055 sessions.

Below (Table 2) we present a sample of the final data. Table 3 -data dictionary and mapping of units.

Table 2: Sample of final -session level dataset

sessionID	Station / space	kWhDelivered	Connected hours
2_39_78_362_2018-04-25	2-39-78-362 / CA-496	7.93	2.20
2_39_95_27_2018-04-25	2-39-95-27 / CA-319	10.01	11.19
2_39_79_380_2018-04-25	2-39-79-380 / CA-489	5.26	9.32

Table 3. Data dictionary -key variables and its usage in each unit (M1 -SARIMAX; M2 -Apriori)

Variable	Type / meaning	Origin / preparation	Used in
sessionID	Text, unique session ID	API field	U1, U3 M2
stationID, spaceID	Text, charging station and space	API field	U1, U2, M2
connectionTime, disconnectTime	Datetime, session start and end	API, converted to local time	U1, U2, M1
kWhDelivered	Number, delivered energy (kWh)	API field, range checked	U1, M2
connected_h	Number, hours plugged in	Derived from timestamps	U1, M2
charging_h	Number, estimated charging active hours	Derived, missing if doneChargingTime invalid	U1, M2
avg_power_kW	Number, average power	Derived 10kW> set to missing and flagged	U1, M2
hour, dayofweek, date	Integer/date, time attributes	Derived from local time	U1, U2, M1
regime	Category: Free, capped, paid, COVID/recovery	Policy dates derived	U1, U2, M1, M2
is_weekend, is_morning_arrival, is_long_stay	Binary, behaviour indicators	Engineered	U2, M2
userID / userInputs	Text/object, User ID and requested stay/energy	API field, used if available only, Pseudonymous	M2

2.2 Data quality, privacy and ethics

Table 4 provides an overview of the quality checks, the issues identified and corrections undertaken.

Table 4: Data quality checks and corrections

Check	Finding	Correction / evidence
Missing values, duplicates	No duplicates. Missing doneChargingTime in 2055 sessions	Retain sessions, flagged done_time_missing; remove only if completion time needed.
Type, range, consistency	28 bad charging end times, 37 sessions more than 72 h connected, 39 sessions more than 10 kW power	Bad end times to missing. Remove extreme durations sessions, logged. Set implausible power to missing and flagged
Units suitability	All required fields exist. Gap of 106 days of recording	Timestamps to local time. Remove the gap days in demand, forecasting analysis.
Synthetic data	No synthetic data used.	Only observed ACN sessions

Finally we have 31387 sessions (99.9% from 31424) retained in a acn_clean_sessions.csv and feature_engineered_sessions.csv, 37 over 72hrs records in acn_flagged_sessions.csv for traceability.

Privacy & ethics: the data is anonymized at the source. Only pseudonymous ids of drivers (no attempt to de-anonymize drivers) and for a small fraction of sessions (user input fields). Team uses data as per research terms of the platform. No publication of IDs other than the pseudonymous ids. API token not included in the repo. For data minimisation (e.g. app-user status for behaviour categories) based on India Digital Personal Data Protection Act 2023. We only use the pseudonymous user fields for behaviour categories.

Biases: Data from 1 dominant (mostly commuter) workplace site -no guarantee of generalization to other sites. User input fields for a subset of sessions-possibility of self-selection. 106 days gap and COVID period -> structural break. Test for bias in predictive model across periods in Section 4.

2.3 Transformations and usable dataset

Table 5 shows transformations that were actually performed with accompanying notebooks.

Table 5. Transformations performed and datasets generated

Step	What was done	Numbers / details
Cleaning (EV_Charging_Demand_Analysis_i9.ipynb)	Remove bad records and extreme stays	31387 of 31424 sessions (99.9%). Remove 37 sessions with >72 h parking. Remove gap in recording (106 days)
Feature engineering	Time zone adjustments. Compute session metrics	Total stay, charging, idle, Avg power (kW).
Daily aggregation	Group by day. Create features for modelling	1104 daily records. (first 28 warm up days excluded for creating lags). 19 features -day of week, lag (1,7,14), rolling averages, holiday, pricing flags.

Cleaning vs model preprocessing. Cleaning (on record level, Table 5) is done once for everything. The model-specific pre-processing differs. For example, lag and rolling features consider only past days. Training data only is used to fit the encoding/scaling for the linear models. Time_series_daily.csv, association_onehot.csv (for SARIMAX and Apriori) are created from the same set of clean sessions. The time-related split into train and test is used for chronological separation (so that no future data is used in training, Section 4).

3. Exploratory Data Analysis and Visualization (Unit 1)

First analytical question is investigated in exploratory analysis. The code written in Python (pandas and Matplotlib) is found in EV_Charging_Demand_Analysis_i10.ipynb, EV_Charging_Demand_Analysis_i11.ipynb. The figures are generated based on project data.

3.1 Demand over time and under different policy regimes

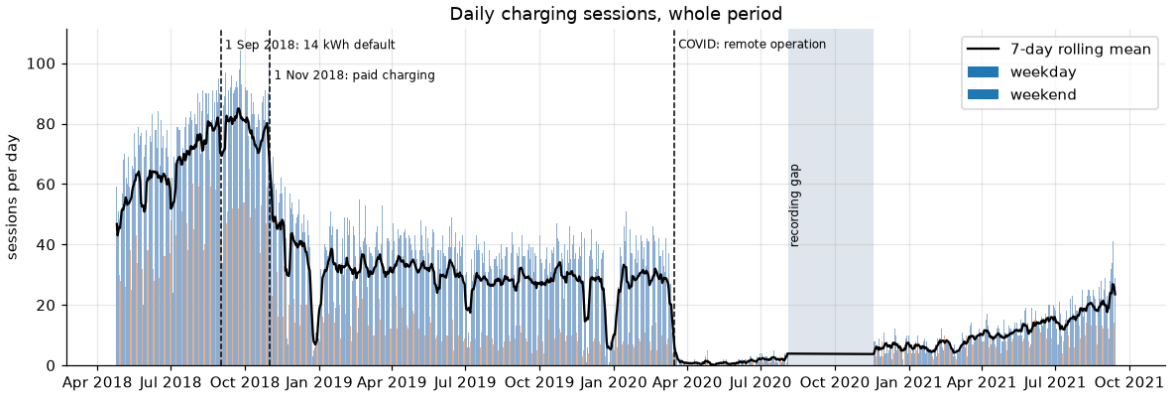


Figure 2. Charging sessions per day for entire period. 7 days rolling mean. (Python/Matplotlib)

Axes: date and sessions per day. Dashed lines show policy regime changes. Shaded area -recording gap. Source: EV_Charging_Demand_Analysis_i11.ipynb

In Figure 2, the demand is shown for 1,132 calendar days, for which data is available. It is possible to see four regimes: Free R1 (74.0 sessions/day on weekdays), Free R2 (88.5 sessions/day on weekdays), Paid R3 (36.8 sessions/day on weekdays), COVID R4 (8.7 sessions/day on weekdays). Thus, the demand level is mainly defined by the policy and operation conditions. So it is necessary to provide regime information when we forecast.

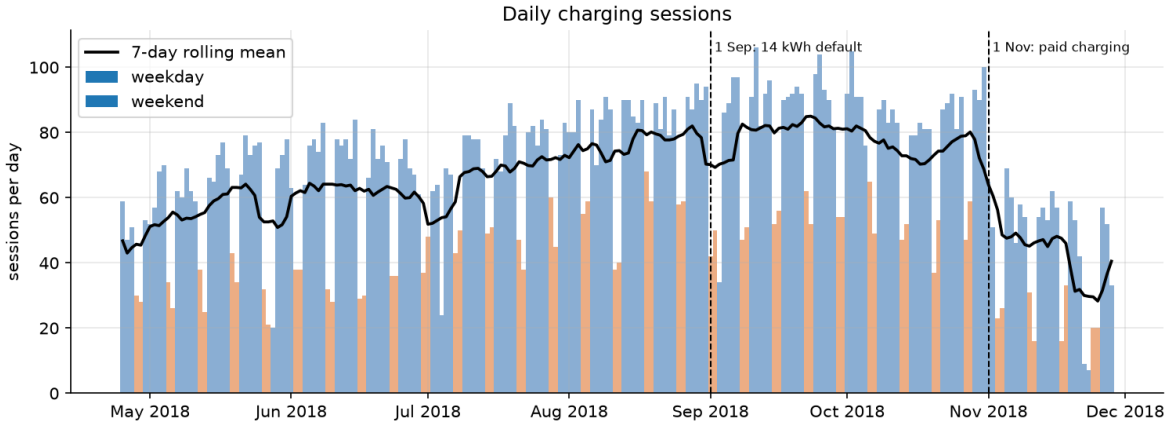


Figure 3. Sessions per day from April till November 2018. Weekdays and weekends. (Python/Matplotlib)

Dashed lines show the changes: 1 Sep 2018 (14 kWh default), 1 Nov 2018 (paid charging)

We can see that when charging switched to paid on 1 Nov 2018, weekday sessions in the subset of Review 1 reduced by 44% (from 89 sessions/day to 50 sessions/day). Or by 58% compared to the entire free period R2. In the final data, there were 54/55 chargers impacted, with evening arrivals decreasing from 19%(R2) to 8% (R3) of weekday arrivals. The transition to a charging for money regime is closely related to the biggest, sustained, decline in demand. Revenue per session needs to be weighed up with charger utilisation for the operator.

3.2 Time-of-day demand/ occupancy

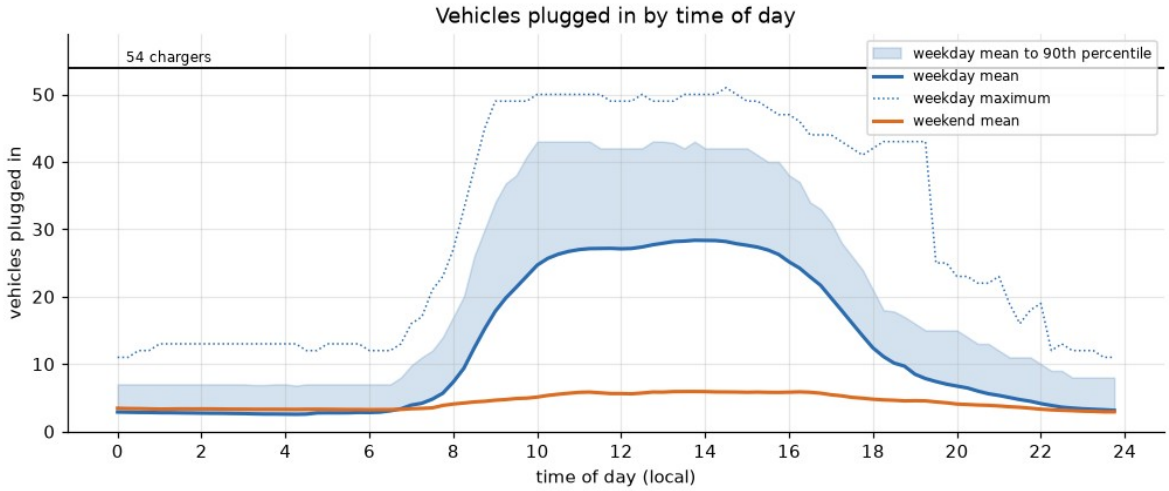


Figure 4. Time-of-day of plugging in (mean of weekday, 90th percentile, max; mean of weekend) (Python/Matplotlib)

Axes: local hour -vehicles plugged in; Horizontal line: 54 chargers. Source: EV_Charging_Demand_Analysis_i11.ipynb

While arrivals are most numerous around 08:00 the occupancy peak is 13:00-14:00. 28.1 vehicles are plugged in on average for the weekday, with a maximum of 51/54 (94% utilisation of the bays). The 5-hour offset between arrivals and occupancy is telling; the "crushing" period is the plateau of late morning to early afternoon, not the arrival itself. With the arrival-between 07:00-10:00 being 45% of weekdays and only 5/493 weekdays being at 90% occupancy, over-capacity is not a common event. So, this data points to the solution not being the installation of more chargers, but rather that of mitigating the peak period.

3.3 Idle time

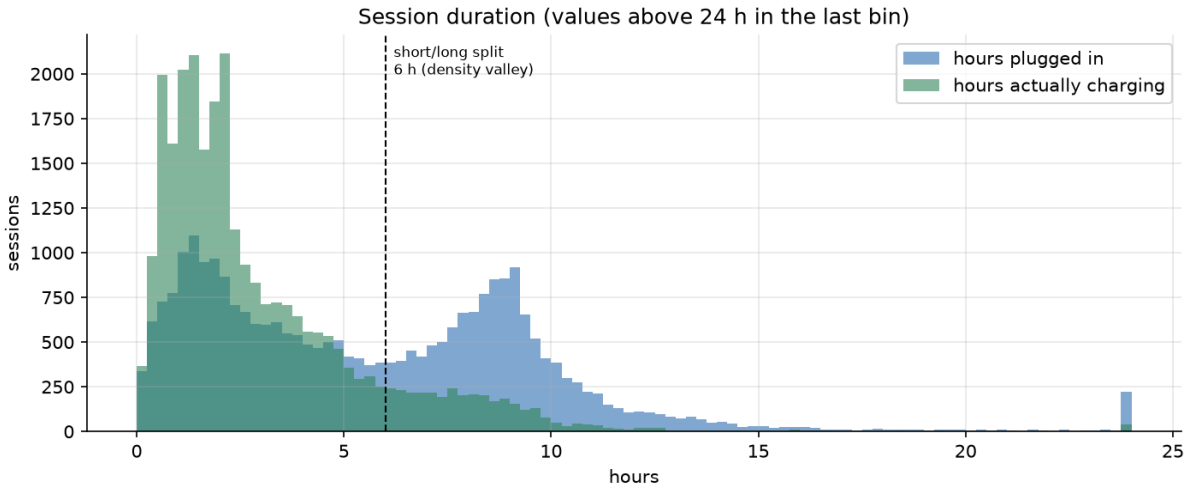


Figure 5. distribution hours plugged in vs hours actually charging (Python/Matplotlib)

Dashed line is 6-hour short/long split at a valley of density. Source: EV_Charging_Demand_Analysis_i11.ipynb

Sessions tend to be long in terms of required charging. Median plugged-in duration is 5.2h while median charging duration is only 2.1h. So 48% of plugging in time is idle. Arrivals in the morning median time plugged in for 8.1h. So the long duration, and regular, would suggest the possibility of smart charging and idle time policy.

What we've learned that we'll use in the modelling. (i) Regime and holiday variables should be part of it. (ii) Day-of-week, recent demand; the weekends are ~58% less sessions. (iii) 6h split of staying, arrival period/energy/charging share bands are what we'll go association rule mining on.

4. Feature Engineering and Predictive Modelling (Unit 2)

Task, target, predictors The task is next day regression of the total daily number of charging sessions for all chargers. We have 1,104 usable daily records. 19 predictors: the day of week (number and 6 weekday dummy), weekend flag, major holiday (campus closure) flag, lags (1, 7, 14), 7 and 14 days rolling mean, 7 days rolling standard deviation, previous 4 same weekdays mean, regime flags (R2 policy, paid, covid) We exclude predictors that are always constant for a given window training set (17-19 window predictors). All column names set to predictor, target or same day outcome (only predictor in models). Lags calculated on a full calendar day. Checked that lag_1 == last day calendar.

Baselines and models We have baselines based on same weekday last week, 4 same weekdays mean, mean for each weekday across all training days (the latter fails after regime changes, eg WAPE 68.8% for W2). Considered models: Ridge, Lasso, ElasticNet (penalised linear) Poisson regression, Random forest, Gradient boosting, Histogram Gradient Boosting (HistGB) -all using scikit-learn [5].

Validation protocol We use strict chronological order in evaluation Test window for 3 regimes: free (31 days, W1), paid (91 days, W2), covid recovery (105 days, W3) In-training data we used cross validation with 5 time order folds. We chose model by minimum error over 5 fold (35.6% WAPE for HistGB)

Leakage audit (i) Fit preprocessing and scaling on training folds only. (ii) No more same day mean or stats on all data for station as predictors (possible leak in previous review) (iii) Lags only last days. (iv) Splits done in time order so no future days in training.

In table 6 we compare models on the same held-out basis.

Table 6. Held-out comparison of baselines and models for next day demand (test window for paid period W2, Dec 2019 -Feb 2020)

Model	WAPE	MAE (sessions)	RMSE (sessions)	Notes
Baseline: last week, same day	32.8%	8.3	12.5	Previous same day of week
Baseline: same day of last 4 weeks average	31.1%	7.9	11.5	Average of last 4 matching weekdays
Random Forest	23.8%	6.1	8.8	Tree ensemble
Histogram Gradient Boosting (chosen)	21.2%	5.4	7.9	Least 5-fold CV error, about 10 points better than baseline
ElasticNet	18.5%	4.7	6.4	Linear comparison model

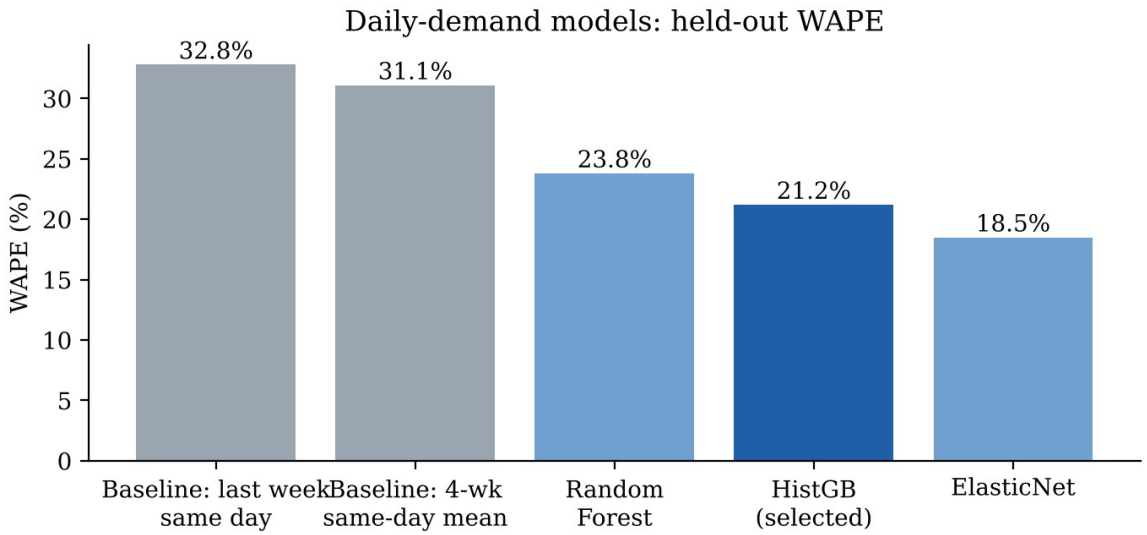


Figure 6. Held out comparison (plot of Table 6) WAPE of baselines and models

We see that all learned models outperform baselines for the paid window (W2). Ridge is the best (17.2% error) followed by Lasso (18.1%) and ElasticNet (18.5%). ElasticNet has the smallest mean WAPE of all 3 windows (15.9%). We choose HistGB as it was the one with smallest 5-fold cross validation error (35.6% WAPE). This one outperforms best baseline for W1 (7.6% vs 8.1%) and W2 (21.2% vs 31.1%) but not the recovery window W3 (32.6% vs 21.4%). So on average HistGB does not consistently outperform simple same-week rules. Nor does it do well for stress windows: first paid month 43.1% vs 42.4%. First COVID weeks 1,214% vs 302%.

Interpretation. We see from permutation importance for HistGB (Figure 7), the most important predictor is day of week by far, followed by yesterday demand (lag_1), 7-day rolling average, and the major holiday flag. None for dummy of each day of week or regime flags in W3. Since we are including lagged demand level in the regime info. Importance is correlation at this site, not a cause-and-effect.

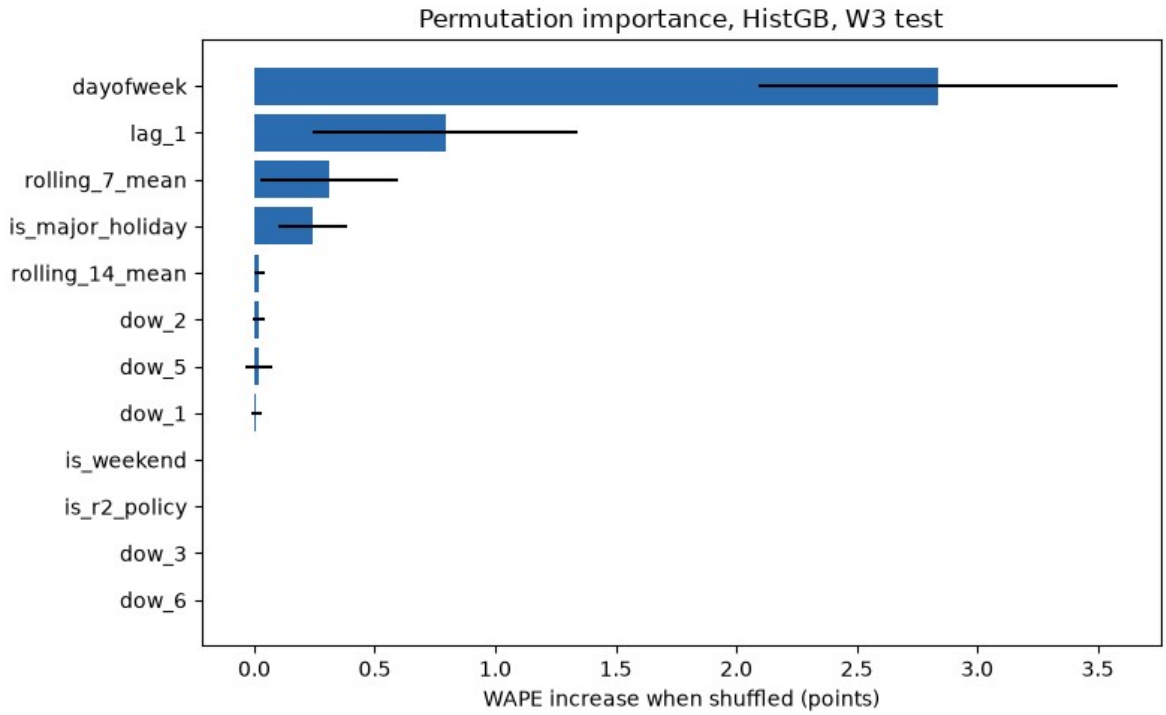


Figure 7. Histogram of permutation importance for HistGB predictors on W3 test window (Python/scikit-learn)

Axes: increasing of WAPE (points) after shuffling each predictor (bars - variation of repeats).

Subgroup & bias check. Figure 8: Average error and MAE relative by weekday for each test window. The free window W1 (about 5-10% in relative error w.r.t. average demand is the smallest one) - free time. Paid window W2 is bigger (14-37%). The biggest one is recovery window W3 (about 26-35%). The error for weekends is more relative in W2. It is good in stable mode and bad in the recovery phase after shocks (under-forecast most days, with negative average error in Figure 8. So, it requires frequent re-trains).

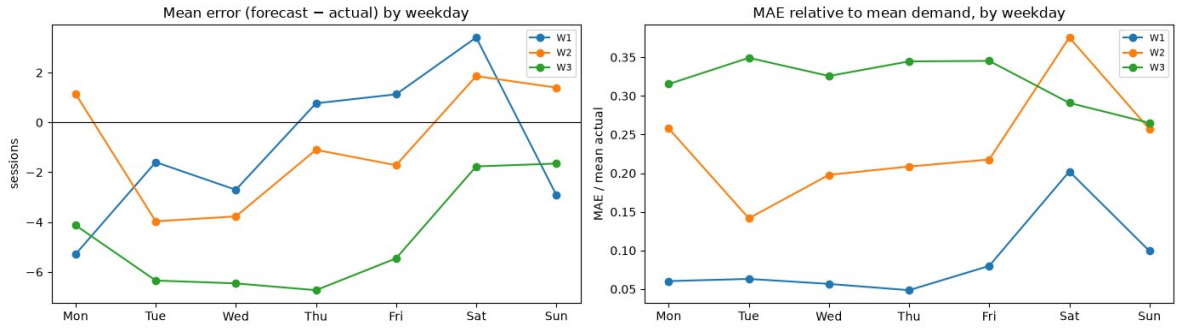


Figure 8. Average error in prediction and relative MAE by weekday for free (W1) and paid (W2) and recovery (W3) windows. (Python/Matplotlib)

Source: EV_Charging_Demand_Analysis_i15.ipynb.

Business suitability. An average 5 sessions per day error for an average day in the paid period is sufficient for grid load planning, maintenance on slow days scheduling, parking management. So, that's enough for the O2a KPI. It is bad in case of abrupt external shocks (like COVID shutdowns). That's why we need the regime aware approach of SARIMAX in section 5.1.

5. Advanced Analytics (Unit 3). Two selected methods

Two methods from Unit 3 are chosen for the business problem, as they address different aspects of the business problem. To answer the question of the expected demand - the Time-Series Analysis (SARIMAX) method. To answer the question of drivers' behavior - Association Rule Mining (Apriori) method.

5.1 Method 1. Time-series analysis (SARIMAX)

Business question. The number of sessions that we expect per day during the next 8 weeks and when peak demand is expected to allow us to plan the number of chargers and maintenance, without overbuying equipment.

Input and preparation. Our input is a time_series_daily.csv file with the daily number of sessions for 1,238 calendar days (1,132 days of recorded data). During 106-day recording interval between 4 Aug and 17 Nov 2020, there was no data (considered as missing rather than zero demand). The data covers 4 different periods of demand: (i) Regime 1 (R1) is free charging. (ii) Regime 2 (R2) is free charging but with 14 kWh default allowance. (iii) Regime 3 (R3) is paid charging ($0.12/kWh). (iv) Regime 4 (R4) is COVID-19 lockdown and post-COVID recovery. As each regime has substantially different demand levels, it would skew our forecast if we combine them together. So we only train the final model for the post-COVID recovery regime (current operation).

Algorithm and settings. We fit a seasonal ARIMA model with weekly seasonal pattern, a campus closure as exogenous variable, and a drift (growth) term. We use Python pandas, statsmodels [6] to fit the model. We refit the model every week. We use 80% and 95% prediction intervals for forecasting.

Evaluation. We use leak-free rolling origin backtest of 28 days ahead forecasts with 16 weeks (448 days of forecasts) cutoffs from Nov 2020 to Sep 2021 to evaluate our model. We use MAE, RMSE and WAPE as error metrics. We provide a Table 7 with the comparison of selected model vs. baseline in the same setting.

Table 7. Backtest of rolling origin for 28 days ahead forecasts (Nov 2020 -Sep 2021).

Approach / configuration	MAE	RMSE	WAPE
A: 8-week mean (weekday) (baseline)	3.86	4.93	25.0%
B: SARIMAX + closures + drift (selected)	3.18	4.08	20.6%

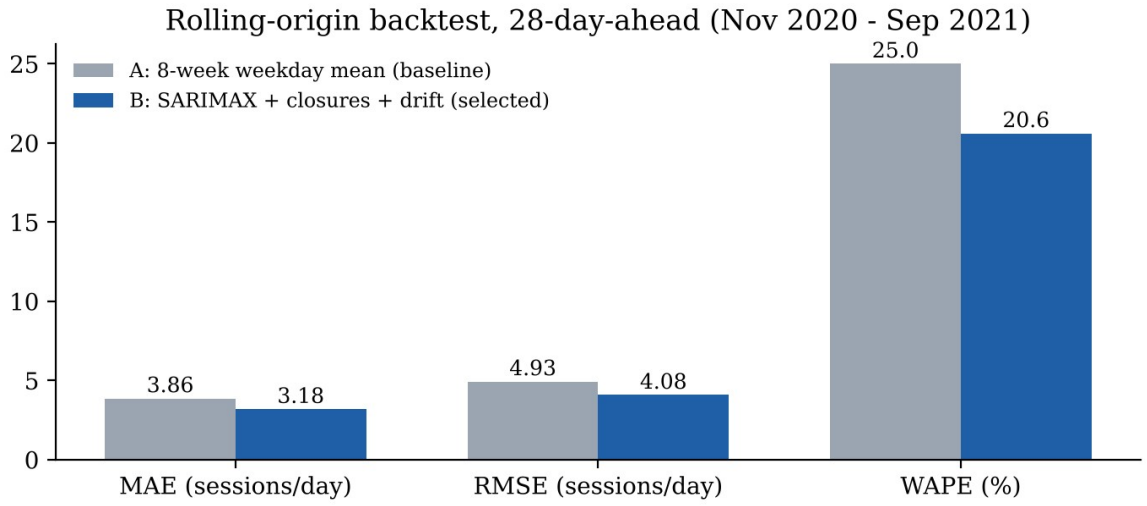


Figure 9. Backtest accuracy for SARIMAX vs weekday mean (baseline)

Results. SARIMAX lowers MAE by 18% compared to the baseline. For the last holdout sample (August-September 2021) it is 3.9 sessions/day MAE and 19% WAPE. 3.18 is the average of all backtest cutoffs. 3.9 is the error on the last, more difficult window. For 8 week ahead forecast, the average number of sessions/day is around 24.7 (28.3 for weekdays). Peak day is Friday 5 Nov. Peak window is 10:30-15:45. Peak around 21 vehicles simultaneously vs. 33 in use chargers.

Interpretation and limitations. The forecasting accuracy is around ±4 sessions per day, enabling capacity planning up to 1-4 weeks ahead. The 80% prediction interval had only 61%

coverage, so use 95% for operational buffers until demand keeps rising. The model is trained on a regime and single site. Assumes future growth at recent pace.

5.2 Method 2 -Association Rule Mining (Apriori)

Business question. What are the combined charging behaviours (arrival time, stay duration, energy, idle time)? to inform charger allocation, idle time policy, etc.

Input/Preparation. Each of 31387 validated sessions is a transaction (association_onehot.csv). 36 items with 11 attributes: arrival period, type of day, duration, energy band, % charging, power, app user, request result, overnight, group of stations, regime. Bin edges based on data (e.g. short/long stay split on 6h, 13-14kWh default allowance). Unreliable: no item.

Algorithm/parameters. Used Python (pandas, numpy), Apriori [4]. Minimum support for mining 1%. Up to 4 items in itemset. Only keep rules with support ≥2% (628 sessions), confidence ≥60%, lift ≥1.2. Also, a longer rule must be at least 5 points more confident than any shorter versions. 9725 frequent itemsets and 35149 candidate rules in total.

Evaluation. Table 8: four configurations are evaluated on the same set of transactions. For each final rule, it is re-evaluated in each regime. If lift ≥ 1.1 then the rule is considered stable within that regime.

Table 8: Comparing configurations of association rules for 31,387 transactions

Approach / configuration	Rules	Kept for interpretation	Stable across regimes
A: all candidates (support ≥ 1%)	35,149	–	–
B: Only thresholds (s ≥ 2% c ≥ 60% lift ≥ 1.2)	3,540	Including trivial and redundant	–
C: Thresholds and dropping trivial / redundant (chosen)	295	165 behaviour (+130 policy-driven)	79 out of 98
D: Mined separately by regime	71 / 80 / 77 / 44	202 not seen by pooling	–

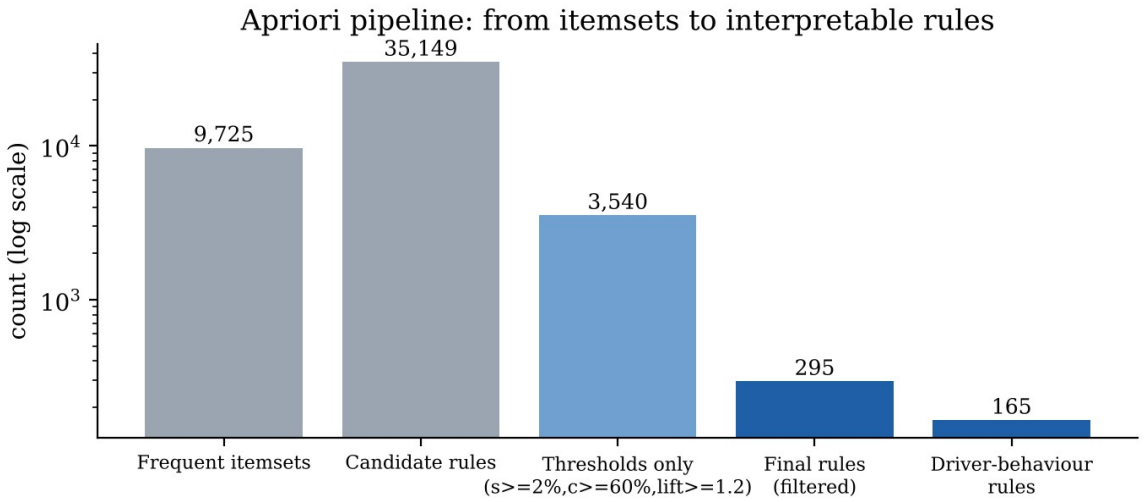


Figure 10: reducing frequent itemsets and candidates to final interpretable rules.

Results and interpretation. We choose config C. We drop 524 true by construction (req → app user), 595 arithmetic (e.g. evening arrival + 10 h → over night), and 1351 redundant rules through filtering. Two exemplary driver-behaviour rules are: arrival in morning, mostly idle → 68% spend 6-10h (lift 2.1) and arrival in evening, mostly charging → 64% spend < 2h (lift 2.4). There are 28% sessions for commuters, but they are 45% and 50% of the charger-time and idle-time respectively. This suggests leverage in idle-time policy.

Limitations. Association is not causality. 130 final rules are operator policy (not driver choice) and we distinguish and report these separately. Rules are dependent on the binning / thresholds.

6. Comparative analysis and business impact

6.1 Two methods of Unit 3 compared

Table 9. Comparison of the two chosen methods for Unit 3

Criterion	SARIMAX (Method 1)	Apriori (Method 2)
Question to answer	Number of sessions / day to expect in the next few weeks (when are the peaks?)	What combinations of behaviours to understand to help with allocation and idle time policies
Input	Daily session counts (recovery regime), closure calendar	31,387 transaction (session level). 36 items
Output	8-weeks forecast + 80% / 95% intervals	295 rules (165 behaviour / 130 policy-driven) Support, confidence, lift
Validation	Roll-origin backtesting MAE / RMSE / WAPE	Compare threshold Stability of lift across regime
Pros	Quantify uncertainty Regime aware >18% baseline MAE	Interpretability Behaviour in demand Stability (79/98 rules)

Criterion	SARIMAX (Method 1)	Apriori (Method 2)
Cons	1 regime 1 site Too narrow in growth	Association -not causation Bin dependent No forecasting

Since MAE is error on forecast and support, confidence, lift is on co-occurrence there is no numerical comparison. We complement these approaches. While unit 2 performs well in a stable regime (~21% WAPE vs 31% baseline) it performs poorly in recovery (~33%) where SARIMAX retrained on the recovery regime outperforms its baseline by 18%. The 10:00-16:00 plateau in occupancy is consistent between unit 1 and forecast. The association rules explain the plateau: long, mostly-idle commuter sessions fill up the chargers during this time period.

6.2 Related studies

Numerical comparison with previous studies cannot be done as they use different sites, time periods, targets, and metrics. Instead we qualitatively compare in Table 10, setting, approaches, and similar results.

Table 10. Related studies comparison

Criterion	Lee et al. [1] (ACN-Data)	Lee et al. [2] (ACN scheduling)	Current project
Data and setting	Descriptive analysis Open workplace session (Caltech, JPL) dataset	Real ACN sessions for scheduling experiments	Our API data 31424 sessions Caltech 2018-2021 4 policy regimes
Approach and validation	Data analysis for demand and user behaviour	Charging scheduling via optimisation	Regime aware ML. SARIMAX chrono backtest. Apriori with stability.
Similar results / limits	Unreliable user reported departure times and long stay.	Leveraging the flexibility from long stay	48% idle plugged-in time and long stay. Including forecasting and behaviour rules. 1 site. No scheduling optimisation

The main novelty in the project is the combination of demand forecasting and behaviour pattern mining in explicit policy regimes based on the data collected and validated by the team. We show that such combination makes sense for the specific case of the Caltech site. No further optimisation scheduling and generalisation over multiple sites are addressed.

6.3 KPI business impact and scenarios

So far, the overall evidence leads to Table 11 implications. This work did not provide the measure of monetary benefit and the operating cost as there is no data in the file with the fields of tariff, revenue or cost. So the table provides capacity/utilisation results. The operator should complete the columns for cost.

Table 11. Business implications linked to KPIs

Section 1 KPI	Evidence (measured or forecast)	Business implication	Status
O1: idle part of the plugged-in time	48% of time plugged in is idle. Commuters account for 28% of sessions but 45% of charger hours and 50% idle hours	Idle fee or move your car notice. Free chargers without hardware	Measured, fee impact not known
O2b: Number of chargers	Required for busiest weekday ≈ 26 with 90% buffer. In use: 33. Installed: 55	No additional chargers. Re-activate idle chargers to match increasing demand	Forecast
O2a: error of day +1	21.2% WAPE vs. 31.1% baseline (for the period being paid)	Plan grid load/maintenance taking into account ±5 sessions	Measured (held-out data)

Scenarios. Table 12 provides the best/base/worst scenarios for the number of chargers in capacity. All under the provided assumptions.

Table 12. Capacity scenarios from the forecast

Scenario	Assumption	Chargers needed	vs 55 installed
Best	The demand will be similar to the current recovery (approx. 24.7 sessions/day). There will be lower occupancy peaks thanks to idle time rules	≈ 21 at peak	Much excess
Base	Forecast the demand with 90% buffer for the busiest weekday	≈ 26	Excess of ≈ 29
Worst	Back to the pre-COVID demand	≈ 37	Excess of ≈ 18

Capacity is enough in all cases, recommendation is resilient to forecast uncertainty. Key factors: recovery in return-to-office demand, use of 95% for buffers, single site assumption. In Section 9 we prioritise recommendations based on impact & feasibility.

7. Interactive Application, Figure Index and Reproducibility

7.1 Mandatory dashboard

Application and access. The EV Charging Analytics is an interactive web dashboard developed using HTML and Chart.js, hosted on Vercel at https://evcharging-alpha.vercel.app/.

Architecture and data flow. The application uses the precomputed output (i.e., cleaned session summaries, forecast holdout, model comparison, association rule) generated by notebooks for reading. Filtering and chart redraw is done in the browser. The Units 1-3 are combined through Overview, Demand, Energy, Forecast, ML Model, Association, Stations tabs. It contains regime/day-type selection for updating KPIs, charts.

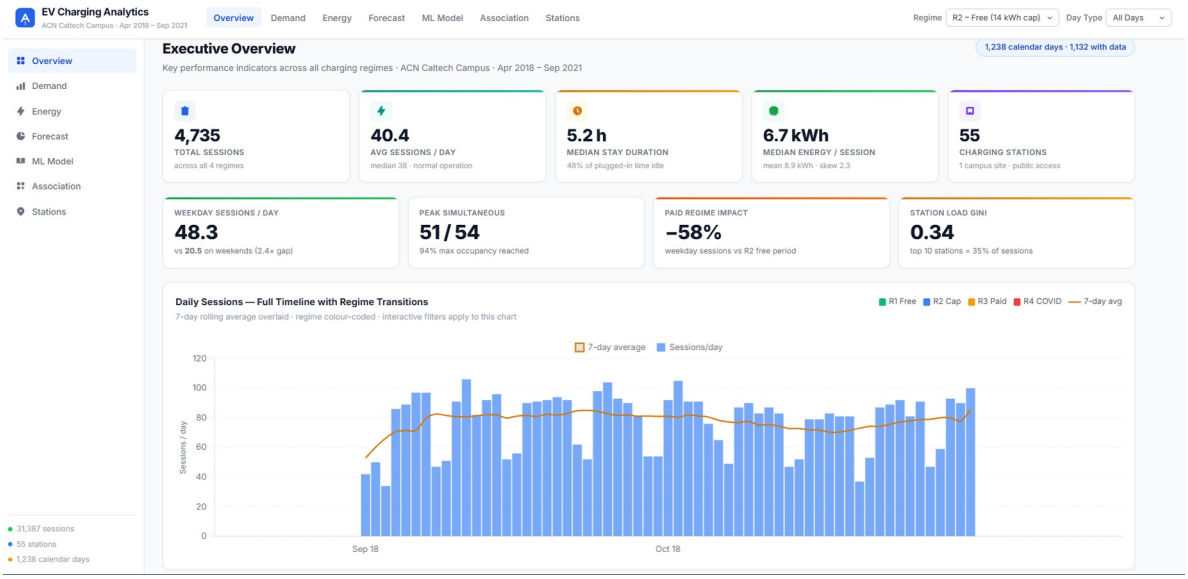


Figure 11. Dashboard Overview. KPI cards, regime/day-type filters, sessions-per-day timeline

Filter state -R2 regime (Free, 14kWh cap). Source: https://evcharging-alpha.vercel.app/

Figure 11: main view with R2 filter. Now that we have charging for pay, we are recovering from demand, do we need more chargers (business scenario): In the Forecast tab. In the next 8 weeks, 24.7 sessions/day expected. Max ~21 cars at once. 33 chargers utilized. No, we don't need more chargers.

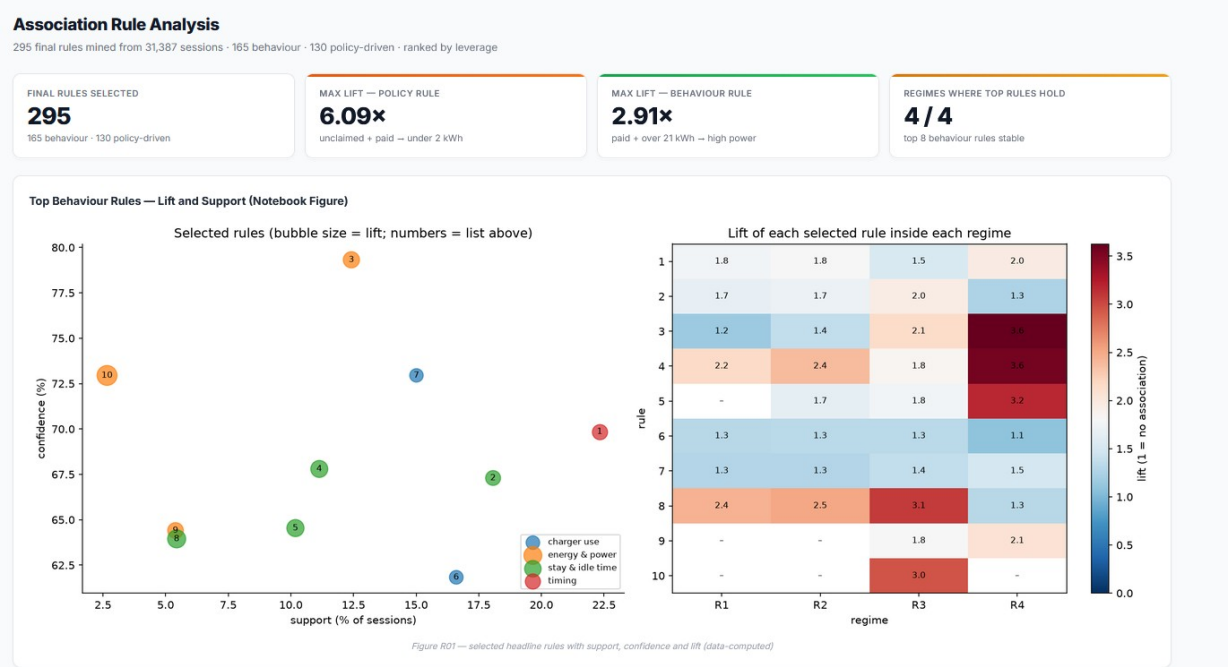


Figure 12. Dashboard Page. Association rule analysis figures. Stability of rules over regimes.

Source: https://evcharging-alpha.vercel.app/

7.2 Figure index and reproduction map

Figures are included only once in Section 1, 3-7. Index in Table 13.

Table 13. Figure index and reproduction map

Figure	Title / tool	Section / key insight	Source notebook / file
Fig. 1	Project workflow / Word table	1.1 - connectivity of units	Report
Fig. 2	Daily sessions full period / Matplotlib	3.1 - Four regimes	EV_Charging_Demand_Analysis_i11.ipynb
Fig. 3	Daily sessions Apr-Nov 2018 / Matplotlib	3.1 - Decrease in paid-charging	…_i11.ipynb
Fig. 4	No. of vehicles plugged-in hourly / Matplotlib	3.2 - 13:00-14:00 occupancy peak	…_i11.ipynb
Fig. 5	Hrs plugged-in v/s Charging / Matplotlib	3.3 - Idle time	…_i11.ipynb
Fig. 6	Model WAPE Comparison / Matplotlib	4 - Model ranking	Plotted from Table 6
Fig. 7	Permutation Importance / scikit learn	4 - Day of week is dominant	…_i15.ipynb
Fig. 8	Error per weekday / window / Matplotlib	4 - Subgroup performance	…_i15.ipynb
Fig. 9	SARIMAX backtesting accuracy / Matplotlib	5.1 - 18% MAE decrease	Plotted from Table 7
Fig. 10	Apriori rules reduction / Matplotlib	5.2 - 35,149 → 295 rules	Plotted from Table 8
Fig. 11	Dashboard screenshot	7.1 - Interactive Filters	Live dashboard
Fig. 12	Association rule dashboard page	7.1 - Lift of rule, regime stability	Live dashboard

7.3 Code and environment

Repository. https://github.com/M-Chandana/EV-Charging-Demand-Analytics.

Structure / Data access. Data/raw, data/processed, notebooks, src, figures, dashboard, reports and README. API token etc. secrets not included. Data are reproduced from ACN-Data API [1, 6] using personal token.

8. End-to-end Validation, Traceability & Contributions

8.1 Completed & Validity

Implementation status. We have implemented and integrated all intended components: data collection & validation, Unit 1 EDA, Unit 2 predictive modelling, both Unit 3 methods, outputs storage and live dashboard. Proof is provided by notebooks, tables and figures indicated earlier.

Threat to validity. Generalisation is limited by single site study. 106 days missing data and COVID structural break affect recovery regime training data. Forecast is not going in wide enough range for increasing demand (80% interval coverage is only 61%) Association rules are just co-occurrence results (operator policy in 130 rules). Only daily totals are forecast. (Complete raw data collected. But figures for Review 1 (Figs 3, 5) are based on earlier partial subset (regenerate figures for complete data if possible)

Table 14. Objective to evidence traceability matrix

Objective / KPI	Analysis / method	Result / metric	Figure / artifact	Recommendation
O1 demand patterns	Section 3 (Unit 1)	Regimes 74.0 / 88.5 / 36.8 / 8.7 weekday sessions/day. 94% peak occupancy	Fig. 2–5	Idle-time policy (Section 9, R2)
O2a next-day forecast	Section 4 (Unit 2)	21.2% WAPE vs 31.1% baseline	Fig. 6–8; Table 6	Same weekday rule after regime change (R1)
O2b multi-week forecast, capacity	Section 5.1 (SARIMAX)	MAE 3.18 vs 3.86. ~26 chargers.	Fig. 9; Table 7	No new chargers. Refit every week. (R1, R3)
O3 behaviour patterns	Section 5.2 (Apriori)	295 rules. 79 stable out of 98	Fig. 10; Table 8	Idle fee/notifications 10-16 hrs (R2)

8.2 Project tracking and individual contributions.

Project tracking. Project board and task history tracked in GitHub Projects Issues. https://github.com/M-Chandana/EV-Charging-Demand-Analytics. Tracking of tasks is done from Backlog To Do In Progress Review/Testing Completed. (Evidence of project board attached in Appendix D) Individual member contribution details in Table 15. Individual marks can differ as per contribution and explanation.

Table 15. Individual contributions.

Team member / register no.	Assigned work	Evidence link / artifact
CB.SC.U4CSE23601 (Adithyaraman R)	Perform time series analysis and forecasting for charging demand	github.com/M-Chandana/EV-Charging-Demand-Analytics/issues/16
CB.SC.U4CSE23228 (Muppidi Chandana)	Analyse and interpret time series forecast results	…/issues/17
CB.SC.U4CSE23612 (D Bhuvaneshwari)	Interactive EV Charging Analytics dashboard	…/tree/main/.github
CB.SC.U4CSE23662 (Khandavilli V V S D Gnanesh)	Association rule mining for charging behaviour	…/issues/6

Shared Unit 1 and Unit 2 deliverables. We have shared deliverables from Unit 1 and Unit 2. All of us have worked together on Review 1 API connection, Paginated Collection (acn_raw_sessions.csv, acn_structured_sessions.csv), Data Cleaning and Validation (EV_Charging_Demand_Analysis_i9.ipynb), Unit 1 Exploratory Analysis (EV_Charging_Demand_Analysis_i10.ipynb, EV_Charging_Demand_Analysis_i11.ipynb), Unit 2 Feature Engineering and Predictive Modelling (EV_Charging_Demand_Analysis_i12.ipynb, EV_Charging_Demand_Analysis_i15.ipynb), final report.

8.3 AI Use & Originality Declarations

Generative AI disclosure. We used generative AI just for language edit, formatting help, debugging help on dashboard and report preparation. All data collection, data validation, coding for modelling, statistical outputs, and conclusion are created by our team and confirmed by team on the project notebooks.

Originality. Our work is original. All ideas/figures/codes which we have borrowed are cited. Turnitin Similarity ≤ 15% (limit 15%). Similarity report is attached here (Appendix D (EV_Charging_Capstone_Final_Report.pdf)).

9. Conclusion, Recommendations and Roadmap

We achieved major part of business objectives as listed in traceability matrix in section 8.1. We characterized demand pattern. Forecasted multi-week demand. Validated against baseline. Mined recurring pattern of behaviour. Results are presented in live dashboard. Partially achieved next day forecast. Our chosen model was better than simple same-weekday rule in steady times but not after covid regime change. Overall, we concluded that there is no need for new chargers at the site. The constraint here is the idle parked car during 10:30-15:45 peak. We also noted that the biggest and lasting reduction in demand happened at the time of switching to paid charging.

Table 16: Prioritised Recommendations.

#	Recommendation	Expected KPI effect	Stakeholder	Feasibility
R1	Week-long, refitted weekly, SARIMAX for 1-4 week capacity planning. Same-weekday rule for day ahead capacity planning post-change. Add buffer on 95% interval	≈ 3.9 sessions/day MAE. Forecast error reduction	Facility operations	High
R2	Introduce idle fee (after grace period) or move-your-car notifications for 10:00-16:00	Idle share reduction (48% currently). Peak occupancy reduction.	Charging operator	Medium (impact unknown)
R3	No new chargers. Re-activate idle chargers as demand increases. Maintenance over weekends, closure days, before 08:00 and after 21:00	No capital expenditures. ~26-37 chargers needed vs 55	Facility management	High

Roadmap: 0-1 month - execute weekly SARIMAX refit (1 analyst, existing notebooks, ACN-Data API token). Dashboard to ops. Success criteria: maintain weekly MAE ~4 sessions/day. 1-6 months - move-your-car notifications (10:00-16:00) for pilot (operator app / email channel. facility staff). Success criteria: idle share, peak occupancy. 6-12 months - idle fee consideration (operator pricing decision. Change in billing system). Re-evaluate activation of chargers vs demand forecast. Success criteria: chargers needed ~ 26-37

Future scope (outside of completed work): Live data refresh. Hourly, per station forecast. Weather / Calendar input. Controlled idle fee trial. Multi-site validation.

References

[1] Zachary J. Lee, Tongxin Li, Steven H. Low. 2019. ACN-Data: Analysis and Applications of an Open EV Charging Dataset. In Proceedings of the 10th ACM International

Conference on Future Energy Systems (e-Energy'19). ACM, New York, NY, USA, 139-149.

[2] Zachary J. Lee et al. 2021. Adaptive Charging Networks: A Framework for Smart Electric Vehicle Charging. IEEE Transactions on Smart Grid 12, 5 (Sept. 2021), 4339–4350.

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[6] ACN-Data. California Institute of Technology. ACN-Data REST API documentation. https://ev.caltech.edu/dataset (accessed 2026).

Appendices

Appendix A -Data documentation. Extended data dictionary (59 engineered variables), validation records (acn_flagged_sessions.csv) and no synthetic data used confirmation.

Appendix B -Additional analysis. Model tuning and error analysis tables for Ridge, Lasso, ElasticNet, Poisson, Random Forest, Gradient Boosting, HistGB.

Appendix C -Implementation. Config, sample output, Dashboard user guide, installation command.

Appendix D -Track and declaration. Project board evidences, individual contribution records, AI declaration, Turnitin similarity report.